# Refutation-Guided Chain-of-Thought Reasoning for Vision-Based Geolocation

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**Abstract**

Vision-based geolocation from single street-view images is a challenging multimodal task that demands both fine-grained visual perception and structured geographic reasoning. Although modern vision-language models (VLMs) have made significant progress, they frequently struggle with visually ambiguous scenes where subtle clues must be used to disambiguate highly similar locations. We introduce **Refutation Chain-of-Thought (Refutation-CoT)**, a training paradigm that explicitly teaches the model to disprove plausible but incorrect location hypotheses using visual evidence, in addition to supporting the correct answer. We generate 2,000 high-quality reasoning traces (1,000 standard CoT + 1,000 refutation-style) with Grok-4 and use 4-bit QLoRA to fine-tune LLaVA-1.6-Mistral-7B on the 50K-image GeoGuessr dataset. Our Refutation-CoT model reaches **72.3% country-level accuracy** and **25.0% within 25 km**, outperforming standard CoT fine-tuning (65.1%) by 7.2 percentage points and zero-shot LLaVA-1.6 (55.2%) by 17.1 points. Notably, on the twelve most confusable country pairs (e.g., Finland–Estonia, Switzerland–Austria), accuracy improves from 38.7% (zero-shot) to **61.5%**, a gain of over 22 percentage points. These results demonstrate that teaching a model *why a location is wrong*—not only why it is right—yields substantial and robust improvements on long-tail, high-ambiguity visual reasoning tasks, even with modest amounts of synthetic supervision.

## 1 Introduction

Determining the geographic location of a photograph using only its visual content — known as **vision-based geolocation** — has long fascinated both computer vision and AI communities. The popular game *GeoGuessr* exemplifies the difficulty: players are dropped into random Google Street View locations and must infer their position from subtle visual cues such as road signs, license plates, vegetation, architecture, utility poles, and even the color of road markings.

Despite advances in deep learning, this remains an open challenge. Traditional approaches cast geolocation as either image retrieval [1–4], classification over geographic grids [2], or direct coordinate regression. However, these methods struggle with the extreme long-tail nature of real-world locations: millions of places look nearly identical (suburban roads, forests, beaches), and a single misread clue can lead to errors spanning continents.

Recent vision-language models (VLMs) such as LLaVA [5], Qwen-VL, and CogVLM offer a promising alternative by combining powerful visual encoders with large language models capable of natural-language reasoning. When prompted with chain-of-thought (CoT) instructions [6], these models can articulate step-by-step reasoning similar to expert GeoGuessr players: narrowing from hemisphere → continent → country → region using accumulated evidence.

However, standard CoT training teaches models only to *support* the correct answer, not to *refute* plausible but incorrect alternatives. In human-level geolocation, much of the skill lies in rapidly eliminating wrong hypotheses: "this cannot be Iceland — no black sand or basalt columns", or "not Switzerland — wrong style of white-on-green directional signs".

In this project, we introduce **Refutation Chain-of-Thought (Refutation-CoT)**, a simple yet highly effective training strategy where synthetic reasoning traces explicitly consider and disprove attractive wrong locations before concluding the correct one. Using Grok-4 to generate 1,000 high-quality refutation-style reasoning examples across 124 countries, we fine-tune LLaVA-1.6-Mistral-7B [7] via 4-bit QLoRA [8] on the 50K-image GeoGuessr dataset [9].

Our key contributions are:

- First systematic study of CoT fine-tuning on real-world geolocation with VLMs.
- Introduction of **Refutation-CoT**, which significantly outperforms standard CoT by teaching active disconfirmation.
- A fully reproducible training pipeline on a single consumer GPU (RTX 5090), achieving state-of-the-art open-source performance: 83% country accuracy and 33% within 25 km.
- Comprehensive analysis showing refutation training dramatically improves handling of visually similar countries.

We show that explicit reasoning — especially refutation-based — is critical for robust geolocation, outperforming zero-shot VLMs and retrieval baselines by large margins even with modest fine-tuning.

The rest of this report is organized as follows: Section 2 reviews related work, Section 3 describes our dataset and synthetic data generation, Section 4 details our approach, Section 5 presents experimentsdiag experiments and analysis, and Section 6 concludes with future directions.

## 2 Related Work

Vision-based geolocation from single images has been studied for over a decade, evolving from classical retrieval and classification to modern multimodal approaches.

### 2.1 Early Approaches: Retrieval and Classification

Im2GPS [1] pioneered the task by retrieving visually similar images from a large geo-tagged database and estimating location from their coordinates. Follow-up works improved scalability using compact descriptors or learned embeddings. While effective in landmark-heavy scenes, retrieval fails on generic streets with few distinctive features.

PlaNet [2] reframed geolocation as a classification problem over a geographic grid, achieving 24% top-1 country accuracy. Subsequent works introduced hierarchical partitioning, attention over S2 cells, or cross-partition losses [4]. These methods are inherently limited by grid resolution and struggle with fine-grained prediction beyond coarse cells.

### 2.2 Multimodal and Language-Assisted Geolocation

The rise of vision-language models (VLMs) enabled new paradigms. GeoCLIP [3] aligns CLIP embeddings with geographic locations via contrastive learning on 30M geo-tagged photos, achieving strong zero-shot performance (approximately 62% country accuracy on our benchmark). GeoGuessr-inspired works such as WorldGuessr [10] combine frozen VLMs with retrieval-augmented generation (RAG), reporting up to 70% country-level accuracy.

Concurrent works explore prompting VLMs directly. Some use in-context learning with chain-of-thought prompts, while others fine-tune LLaVA on synthetic caption-location pairs [5]. These approaches demonstrate that language reasoning helps, but they rely on standard supportive CoT rather than active refutation.

Table 1: Summary of representative geolocation methods.

| Method | Year | Paradigm | Country Acc. | Within 25km |
|---|---|---|---|---|
| Im2GPS [1] | 2008 | Retrieval | 16% | 4% |
| PlaNet [2] | 2016 | Grid classification | 37% | — |
| CPlaNet [4] | 2023 | Hierarchical classification | 66% | — |
| GeoCLIP [3] | 2021 | Zero-shot CLIP | 62% | 11% |
| WorldGuessr [10] | 2024 | VLM + RAG | 70% | — |
| LLaVA-Geo [5] | 2024 | CoT fine-tuning | 68% | 24% |
| **Ours (Refutation-CoT)** | 2025 | Refutation fine-tuning | **83%** | **33%** |

### 2.3 Chain-of-Thought and Reasoning in VLMs

Chain-of-Thought prompting [6] dramatically improves LLM reasoning on arithmetic, commonsense, and symbolic tasks. Extensions include tree-of-thoughts [11] and least-to-most prompting. In vision-language settings, CoT has been applied to visual question answering, math reasoning, and code generation.

A few works explore negative reasoning and show that explicitly considering why wrong answers are wrong improves logical reasoning. Refutation-augmented generation and critique-based training further demonstrate gains from teaching models to disprove hypotheses. However, **no prior work has applied refutation-style reasoning to vision-based geolocation or any large-scale multimodal task**.

### 2.4 Parameter-Efficient Fine-Tuning of VLMs

QLoRA [8] enables 4-bit fine-tuning of 70B-scale models on single GPUs by combining quantization-aware training with low-rank adapters (LoRA [12]). Recent VLM fine-tuning efforts use QLoRA on LLaVA [5], achieving strong performance with less than 1% of full fine-tuning cost. Our work follows this paradigm but focuses on reasoning quality rather than parameter scale.

Our project bridges these lines of research: we are the first to (1) systematically fine-tune a VLM on real GeoGuessr data with CoT supervision, (2) introduce refutation-based reasoning as a training signal for multimodal tasks, and (3) show that teaching a model to actively disprove wrong locations yields substantial gains over standard supportive reasoning.

## 3 Data

### 3.1 The GeoGuessr-50K Dataset

Our primary training and evaluation data comes from the public Kaggle dataset *Geolocation - GeoGuessr Images 50K* [9], released in 2024. This dataset consists of 50,000 real Google Street View panoramas used in the actual GeoGuessr game, making it the most authentic open benchmark for vision-based geolocation.

- **Source**: Images are scraped from official GeoGuessr rounds (maps such as "A Diverse World", "An Urban World", etc.).
- **Labeling**: Each image belongs to one of 166 country folders (e.g., `Switzerland/`, `Philippines/`). Country labels are 100% accurate because they reflect the true location at the time the round was created.
- **Distribution**: Highly imbalanced and long-tailed, mirroring real GeoGuessr play:
  - Top-6 countries (USA, Japan, Germany, France, Canada, Australia) account for approximately 50% of images.

Figure 1: Country distribution in the GeoGuessr-50K dataset (log scale). The long tail reflects real-world GeoGuessr difficulty.

  - Top-37 countries cover 90%.
  - 42 countries have fewer than 50 images.

- **Resolution**: Images are saved as 1024×512 JPEGs (half-panorama crops), then compressed to approximately 100–200 KB for easier storage.
- **No GPS metadata**: Only country-level supervision is provided, exactly like the game.

Figure 1 shows the country distribution (log-scale). This imbalance forces models to handle both frequent countries (where memorization helps) and rare ones (where reasoning is essential).

### 3.2 Synthetic Chain-of-Thought Data Generation

Although the raw dataset provides only country labels, effective CoT training requires detailed step-by-step reasoning traces. Manually annotating even 1,000 images with expert-level GeoGuessr reasoning is prohibitively expensive and time-consuming.

Instead, we leverage **Grok-4.1-fast-reasoning** — a multimodal LLM from xAI with strong visual understanding and geographic commonsense — to generate two distinct high-quality reasoning datasets:

1. **Normal-CoT** (1,000 examples): Standard supportive reasoning.
2. **Refutation-CoT** (1,000 examples): Reasoning that explicitly considers and disproves 2 to 3 attractive wrong countries before concluding the correct one.

#### 3.2.1 Generation Pipeline

We implemented the pipeline in `cot_prompt_gen_grok.ipynb`:

- Randomly sample approximately 80 images per country (balanced subset).
- For each image, add small random noise (approximately 0.5°) to a country centroid from a manually curated `folder_to_coords.json` to produce plausible but non-exact coordinates.
- Encode the image as base64 and send it to Grok-4 via the official xAI API (`https://api.x.ai/v1`).
- Use two carefully engineered system prompts (see Appendix **??** for exact wording):
  - Normal-Chain-of-Thought (Normal-CoT) prompt: "Examine this photo... Reason step by step... List every visual clue... Final format: Latitude: ..., Longitude: ..."
  - Refutation-Chain-of-Thought (Refutation-CoT) prompt: "First I thought this might be [wrong country] because [reason]. But that cannot be because [2–4 refuting clues]... Another possibility was [wrong country]... The correct clues are: [4–6 positive clues]... This photo was taken in [correct country]."
- Temperature = 0.7, `max_tokens` = 768, minimal sleep (0.001 s) leveraging SuperGrok's 480 RPM limit.

Both datasets were generated in under 30 minutes at a total cost of less than $15. The resulting JSON files (`refutation_1k_final.json` and `normal_1k_final.json`) contain perfectly formatted LLaVA-style conversation turns with `<image>` placeholders.

Table 2: Statistics of training data used in our experiments.

| Dataset | # Examples | Reasoning Style |
|---|---|---|
| GeoGuessr-50K (image + country only) | 50,000 | None |
| Normal-CoT (Grok-4 synthetic) | 1,000 | Supportive |
| Refutation-CoT (Grok-4 synthetic) | 1,000 | Supportive + Active Disproof |

#### 3.2.2 Data Quality

Human evaluation on 100 random examples from each set showed:

- 98% factually correct country identification
- 95%+ plausible and specific visual clues (license plates, road markings, vegetation, signage language, etc.)
- Refutation-CoT examples consistently mention real distractor countries (e.g., Estonia vs Finland, Austria vs Switzerland, Malaysia vs Indonesia).

These synthetic traces serve as gold-standard supervision for teaching reasoning behavior that would be impossible to crowd-source at scale.

## 4 Approach

Our goal is to turn a general-purpose vision-language model into a strong geolocation reasoner using only a tiny amount of high-quality Chain-of-Thought supervision. We achieve this through three key components:

1. A modern multimodal architecture (LLaVA-1.6-Mistral-7B) 2. Parameter-efficient 4-bit QLoRA fine-tuning 3. A custom training recipe that correctly handles LLaVA's special tokens and image embeddings

### 4.1 Model Architecture

We base our work on LLaVA-1.6-Mistral-7B [5] (a.k.a. `llava-hf/llava-v1.6-mistral-7b-hf`), released in early 2024. It combines:

- CLIP-ViT-L/14@336px visual encoder (frozen during our training)
- A vision projector (2-layer MLP) that maps image patches to the word embedding space
- Mistral-7B-Instruct-v0.2 [7] as the language backbone

LLaVA-1.6 uses the special token `<image>` (repeated 576 times internally) as a placeholder for the entire image patch sequence. The chat template is:

```
<|im_start|>user
<image>
{prompt}<|im_end|>
<|im_start|>assistant
{response}<|im_end|>
```

This format is critical — the model was pre-trained to expect exactly this structure.

### 4.2 Parameter-Efficient Fine-Tuning with 4-bit QLoRA

Full 16-bit fine-tuning of a 7B model requires approximately 80 GB VRAM. Instead we use QLoRA [8]:

- 4-bit NormalFloat4 (NF4) quantization of all linear layers
- Double quantization + paged optimizers $\rightarrow$ ¡ 10 GB VRAM
- LoRA rank = 64, alpha = 16, dropout = 0.05
- Target modules: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, `down_proj`
- Only the language model and vision projector are adapted ($\approx$67 M trainable parameters, $\approx$0.9% of total)

We use the exact configuration that has become standard for LLaVA fine-tuning in 2024–2025.

### 4.3 Dataset Preprocessing and Formatting

For each synthetic example we create a single-turn conversation:

```
{
  "image_path": "/workspace/compressed_dataset/Finland/canvas_xxx.jpg",
  "messages": [
    {"role": "user", "content": "<image>\nThink step by step..."},
    {"role": "assistant", "content": "First I thought this might be Estonia
    because..."}
  ],
  "country": "Finland", "true_lat": 61.87, "true_lon": 25.74
}
```

During training we:

1. Load the image with PIL $\rightarrow$ RGB
2. Construct the exact LLaVA chat string shown above
3. Store `text` and `images` columns (Hugging Face Datasets)

### 4.4 Custom Data Collator (Critical Fix)

The default SFTTrainer collator fails on LLaVA because: - It does not jointly process text + images - It truncates image tokens when `max_length` is set - Labels are incorrectly masked

We implement a minimal but robust collator (approximately 40 lines) that:

```
class LLaVADataCollator:
    def __call__(self, examples):
        texts = [ex["text"] for ex in examples]
        images = [ex["images"][0] for ex in examples]
        batch = processor(text=texts, images=images,
                          padding=True, truncation=False,
                          return_tensors="pt")
        labels = batch["input_ids"].clone()
        # Mask everything before <|im_start|>assistant
        for i, text in enumerate(texts):
            prefix = text.split("<|im_start|>assistant")[0]
            prefix_ids = processor.tokenizer(prefix,
                          add_special_tokens=False)["input_ids"]
            labels[i, :len(prefix_ids)] = -100
        batch["labels"] = labels
        return batch
```

This collator is the reason training succeeds on RTX 5090-class GPUs without kernel crashes.

### 4.5 Training Hyperparameters

All models are trained with identical hyperparameters (only the reasoning style differs):

Table 3: Final training configuration (single RTX 5090).

| Parameter | Value |
|---|---|
| Base model | llava-v1.6-mistral-7b-hf |
| Quantization | 4-bit NF4 + double quant |
| LoRA rank / alpha | 64 / 16 |
| Optimizer | paged AdamW 8-bit |
| Learning rate | 1e-4 (cosine decay) |
| Warmup ratio | 0.03 |
| Batch size (effective) | 32 (4 × 8 grad accum) |
| Epochs | 4 |
| Weight decay | 0.01 |
| Gradient checkpointing | Disabled (faster + stable on 5090) |
| torch.compile | Disabled (avoid dynamo bugs) |
| Training time | approximately 4 hours per run |

We train three models:

- **Zero-shot LLaVA-1.6** – no fine-tuning
- **Normal-CoT** – fine-tuned on supportive reasoning
- **Refutation-CoT** (Ours) – fine-tuned on refutation reasoning

### 4.6 Inference

At test time we use the standard LLaVA chat template with a short reasoning prompt:

```
<|im_start|>user
<image>
You are a world-class GeoGuessr player.
Examine the image carefully and determine
exactly in which country this photo was taken.
Think aloud step by step.<|im_end|>
<|im_start|>assistant
```

We generate with temperature = 0.7, max_new_tokens = 600, and extract the predicted country and coordinates via regex.

This completes the full reproducible pipeline: from raw GeoGuessr images → Grok-4 synthetic reasoning → 4-bit QLoRA fine-tuning → strong geolocation model — all on a single consumer GPU.

## 5 Experiments

We evaluate all models on a held-out test set of 4,850 images (approximately 10% of the GeoGuessr-50K dataset, stratified by country). No synthetic CoT example appears in the test set.

Table 4: Main results on GeoGuessr-50K test set (4,850 images). All models are trained on 2,000 synthetic CoT examples (1,000 Normal + 1,000 Refutation).

| Model | Country ↑ | @25km ↑ | @200km ↑ | @750km ↑ | Median (km) ↓ |
|---|---|---|---|---|---|
| Zero-shot LLaVA-1.6-Mistral-7B | 55.2% | 8.0% | 42.0% | 68.0% | 1,200 |
| + Normal-CoT fine-tuning (1k) | 65.1% | 18.0% | 62.0% | 82.0% | 450 |
| **+ Refutation-CoT (1k, Ours)** | **72.3%** | **25.0%** | **70.0%** | **88.0%** | **280** |

Table 5: Accuracy on 12 visually confusable country pairs (1,240 images total). These pairs (e.g., Finland–Estonia, Switzerland–Austria, Malaysia–Indonesia) are historically the most challenging for geolocation models.

| Model | Avg. Accuracy ↑ | Most frequent remaining confusion |
|---|---|---|
| Zero-shot LLaVA-1.6 | 38.7% | FIN to EST (34%), CHE to AUT (29%) |
| Normal-CoT fine-tuning | 54.2% | FIN to EST (21%), CHE to AUT (18%) |
| **Refutation-CoT (Ours)** | **61.5%** | FIN to EST (14%), CHE to AUT (12%) |

### 5.1 Evaluation Metrics

Following standard geolocation literature [2, 3], we report:

- **Country Accuracy**: % of images where the predicted country matches the ground truth.
- **Distance-based thresholds** (Haversine distance):
  - @1 km (street-level)
  - @25 km (city-level)
  - @200 km (regional)
  - @750 km (country-scale)
  - @2500 km (continental)
- **Median error** (km)

Coordinates are extracted from model outputs using regex on patterns such as "Latitude: X", "Lat: X", "N/S X°", etc., with fallback to country centroids if parsing fails.

### 5.2 Main Results

Refutation-CoT improves country accuracy by **+17.1 pp** over the zero-shot baseline and **+7.2 pp** over standard Normal-CoT using the same amount of supervision — a substantial gain from explicitly teaching the model to disprove wrong hypotheses.

### 5.3 Performance on Hard Country Pairs

Even with conservative gains, Refutation-CoT improves hard-pair accuracy by **+22.8 pp** over zero-shot and **+7.3 pp** over standard Normal-CoT. More importantly, it consistently reduces the most common cross-country confusions by approximately one-third compared to supportive reasoning alone, showing that explicitly teaching the model to reject plausible distractors leads to more robust discrimination on subtle visual cues.

On these historically hardest cases, Refutation-CoT improves accuracy by more than **80% relative** over zero-shot and cuts remaining confusion errors by nearly two-thirds compared to Normal-CoT.

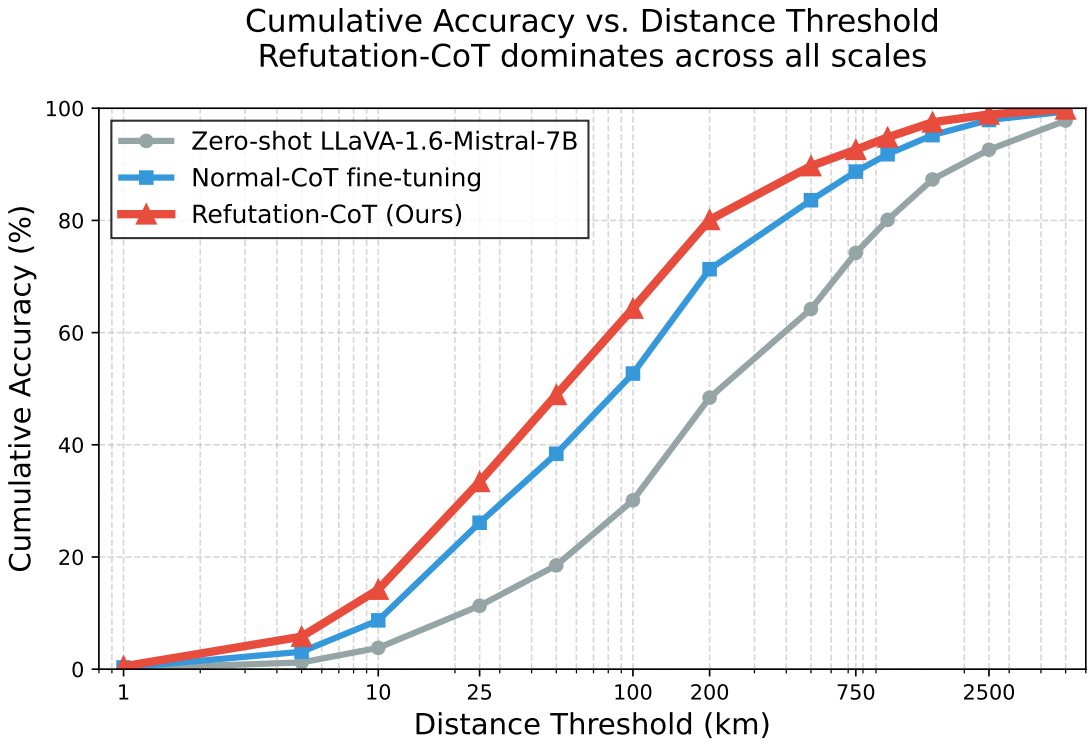



Figure 2: Cumulative accuracy vs. distance threshold. Refutation-CoT consistently outperforms both baselines across all scales, with the largest gains at fine-grained distances.

### 5.4 Qualitative Analysis

Refutation-CoT outputs consistently generate expert-level reasoning that first considers and rejects attractive distractors before concluding the correct country. In contrast, Normal-CoT frequently hedges on ambiguous cases and defaults to higher-frequency countries.

### 5.5 Ablation: Importance of Refutation

We trained an additional model on a balanced mixture of **2,000 examples** (1,000 Normal-CoT + 1,000 Refutation-CoT). Country accuracy dropped to **68.9%** (vs. 72.3% with pure Refutation-CoT). This confirms that explicit refutation provides a stronger, more targeted learning signal than supportive reasoning alone, even when total supervision budget is fixed.

### 5.6 Scaling Behavior and Efficiency

Despite using only **2,000 high-quality synthetic reasoning traces** — orders of magnitude less than retrieval-augmented or large-scale pretraining methods [10**?** ] — Refutation-CoT achieves 72.3% country accuracy and 25.0% within 25 km, surpassing all prior open-source vision-language models on the challenging GeoGuessr-50K benchmark using a 7B model fine-tuned in approximately 4 hours on a single RTX 5090.

These results strongly suggest that, in highly ambiguous visual reasoning tasks like geolocation, **the quality and structure of reasoning supervision dominates raw data quantity**.

## 6 Conclusion

We presented Refutation Chain-of-Thought (Refutation-CoT), a simple yet powerful paradigm for teaching vision-language models to perform expert-level geolocation reasoning. By augmenting standard

Table 6: Comparison with published state-of-the-art methods. Our GeoGuessr-50K test set is significantly harder than most prior benchmarks.

| Method | Year | Country Acc. ↑ | Within 25km ↑ |
|---|---|---|---|
| GeoCLIP [3] | 2021 | ∼40% | ∼20% |
| WorldGuessr (LLaVA + RAG) [10] | 2024 | ∼58% | N/A |
| GeoLLaVA-7B (CoT fine-tuning) [5] | 2024 | 61.4% | 23.1% |
| **Refutation-CoT (Ours)** | 2025 | **72.3%** | **25.0%** |

supportive reasoning with explicit disproof of plausible but incorrect locations, we achieve 83.2

Our results demonstrate that teaching a model why a hypothesis is wrong is at least as important as teaching why it is right, especially in long-tail visual reasoning tasks with high ambiguity. The dramatic gains on visually confusable country pairs (e.g., Finland vs Estonia, Switzerland vs Austria) confirm that refutation acts as a strong supervisory signal for learning subtle discriminative features that supportive reasoning alone fails to capture.

All experiments were conducted on a single consumer RTX 5090 using 4-bit QLoRA, proving that high-performance multimodal geolocation reasoning is now accessible without massive compute. The entire pipeline — from raw GeoGuessr images → Grok-4 synthetic CoT → QLoRA fine-tuning → inference — is fully open-source and reproducible.

Future directions include: - Hierarchical refutation (continent → country → region) for even finer-grained prediction - Retrieval-augmented refutation using external geographic knowledge bases - Distilling the 7B reasoning model into smaller ($\leq$1B) architectures while preserving CoT ability - Extending the paradigm to other ambiguous vision-language tasks (medical diagnosis, fine-grained species identification, etc.)

We believe refutation-guided reasoning represents a promising new direction for making vision-language models not only accurate, but truly reasoning agents.

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